Progressive Electricity Billing

Complete worked solution

Sample tariff: first 100 kWh at 5.00/unit, next 100 at 8.00, and usage above 200 at 12.00. Add a fixed charge of 150, a 5% surcharge on energy charges, and a 3% rebate on the subtotal before surcharge when usage is <= 100. For multiple months, flag an increase over 30% from the previous month (provided the previous bill is positive).

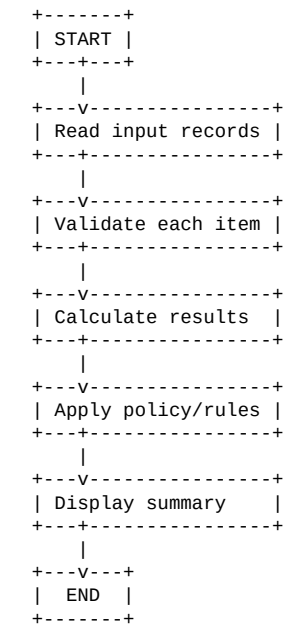
1. Computational Thinking and Algorithm

Inputs: Supplied records and policy values. **Processing:** Validate data, calculate charges or scores, apply ordered rules, and compare outcomes. **Outputs:** Itemized results, classifications, rankings, or recommendations.

Pseudocode

```
START
  Read monthly consumption values
  FOR each month
    Validate consumption >= 0
    Calculate charge in each progressive slab
    Add fixed charge; calculate surcharge and any rebate
    Store total bill in monthly list
    Compare with previous month; flag increase > 30%
  Display monthly itemization, bills, and flags
END
```

2. Flowchart



3. Python Program

The program uses basic variables, numeric and string data, tuples, lists, loops, conditionals, arithmetic, and formatted output. Policy values are constants near the top of the program.

```
months = [("Jan", 85), ("Feb", 120), ("Mar", 230), ("Apr", 100)]
fixed_charge = 150.0
surcharge_rate = 0.05
rebate_rate = 0.03
previous_bill = -1
bills = []

print("Month  Units  Energy  Surcharge  Rebate  Total  Alert")
```

```
for month, units in months:
    if units < 0:
        print(month, "invalid consumption")
        continue
    if units <= 100:
        energy = units * 5.0
    elif units <= 200:
        energy = 100 * 5.0 + (units - 100) * 8.0
    else:
        energy = 100 * 5.0 + 100 * 8.0 + (units - 200) * 12.0
    subtotal = energy + fixed_charge
    surcharge = energy * surcharge_rate
    rebate = subtotal * rebate_rate if units <= 100 else 0
    total = subtotal + surcharge - rebate
    alert = ""
    if previous_bill > 0 and total > previous_bill * 1.30:
        alert = "UNUSUAL INCREASE"
    bills.append((month, total))
    print(month, units, round(energy, 2), round(surcharge, 2),
          round(rebate, 2), round(total, 2), alert)
    previous_bill = total

print("Bills:", bills)
if len(bills) > 0:
    largest = bills[0]
    for bill in bills:
        if bill[1] > largest[1]:
            largest = bill
    print("Highest bill:", largest[0], round(largest[1], 2))
```

4. Sample run guidance

Run the program as written to see a sample result. Replace the sample lists and constants with the values provided in your assignment. Keep the rule order and units consistent; for example, percentages should be entered as percentages and converted to fractions only in the calculation.